

# ECE 565 - DATA MINING

## Weekly Progress Report Template

**Deadline: Every Tuesday by 11:59 PM**

Spring 2026

Instructor: Asst. Prof. Dr. Asli Eyecioglu Ozmutlu

### INSTRUCTIONS

1. Fill out this report every week with your project progress.
2. Submit as PDF (StudentID\_Surname\_Week#.pdf) on Canvas by Tuesday 11:59 PM.
3. Include actual results (numbers, tables, figures) — not just descriptions of what you plan to do.
4. Compare your results against the papers from your proposal. Use the comparison table provided.
5. Be honest about problems. If something didn't work, explain why and what you tried.
- 6. The 'Next Week Plan' section is your commitment — I will check it against the following week's report.**

## WEEKLY PROGRESS REPORT

|                       |                                                                                                                                                      |                    |             |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------|
| <b>Student Name:</b>  | Resul Ozkale                                                                                                                                         | <b>Student ID:</b> | 2540041007  |
| <b>Week Number:</b>   | Week 12 - Final Evaluation and Decision-Policy Revision                                                                                              | <b>Date:</b>       | May 4, 2026 |
| <b>Project Title:</b> | SCRE-Credit: Stability- and Cost-Regularized Ensemble for Credit Risk                                                                                |                    |             |
| <b>Dataset:</b>       | Primary: UCI Default of Credit Card Clients / Taiwan. External validation: FICO HELOC. German Credit / Option A is archived and out of active scope. |                    |             |

## SECTION 1: Work Completed This Week

Describe concretely what you accomplished this week. Use the checklist below and provide details for each completed item.

| ?    | Task                           | Details / Evidence                                                                                                                                                                                                                                                                         |
|------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Done | Data loading & exploration     | No scope change this week. Taiwan remains the primary dataset and HELOC remains the external validation dataset. The active project still excludes German Credit / Option A.                                                                                                               |
| Done | Data cleaning / preprocessing  | Existing train-only preprocessing and 60/20/20 split policy were retained. Leakage guardrails remain active; the final decision summary records that test data is used only for final evaluation evidence.                                                                                 |
| Done | Feature engineering review     | Feature dictionaries remain target-free and row-wise. Taiwan engineered features mainly summarize payment delay, bill/payment behavior, utilization proxy, and volatility. HELOC engineered features summarize burden, delinquency intensity, credit-history length, and inquiry pressure. |
| Done | Hyperparameter tuning          | The Week 11 pending HELOC Optuna task is now completed. Taiwan: 100/100 trials. HELOC: 50/50 trials. Both use 5x3 repeated stratified CV.                                                                                                                                                  |
| Done | Final evaluation & metrics     | Generated final Taiwan and HELOC leaderboards with accuracy, precision, recall, F1, specificity, ROC-AUC, PR-AUC, Brier, ECE, confusion-matrix counts, and FN=5 / FP=1 expected cost.                                                                                                      |
| Done | SCRE-Credit ablation           | Added ablations for full SCRE, performance-only weights, without calibration, without cost, without stability, without faithfulness, without engineered features, and without monotonic models.                                                                                            |
| Done | Explainability and reliability | Computed 50-seed LightGBM SHAP stability protocol and faithfulness diagnostics. I will report this as LightGBM stability evidence only, not as direct CatBoost or SCRE ensemble stability.                                                                                                 |
| Done | Claim control                  | Generated claim_control_matrix.csv. It explicitly prevents unsupported claims such as ?SCRE-Credit outperforms all individual models.?                                                                                                                                                     |
| Done | Reproducibility check          | pytest was rerun in the current project state: 39 passed.                                                                                                                                                                                                                                  |

## SECTION 1b: Show Your Work

**Paste or attach at least one piece of evidence from this week: a code output, a plot, a table of results, a screenshot, or a confusion matrix. Do NOT submit a report with only text and no evidence.**

| Concrete evidence from current artifacts | Result                                                                                                     | Comment                                                                                                      |
|------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Taiwan operational winner                | CatBoost: Acc 0.627, Precision 0.351, Recall 0.809, F1 0.490, ROC-AUC 0.785, PR-AUC 0.564, Cost 3247       | Best expected-cost model under FN=5, FP=1, but precision is weak.                                            |
| HELOC operational winner                 | Scorecard / WOE: Acc 0.587, Precision 0.559, Recall 0.982, F1 0.712, ROC-AUC 0.801, PR-AUC 0.804, Cost 892 | Most interpretable and lowest-cost HELOC result. Specificity is low, so it is not a perfect decision model.  |
| SCRE-Credit Taiwan                       | Acc 0.654, Precision 0.364, Recall 0.756, F1 0.492, ROC-AUC 0.782, PR-AUC 0.544, Cost 3370                 | Competitive framework, but not Taiwan cost winner.                                                           |
| SCRE-Credit HELOC                        | Acc 0.646, Precision 0.601, Recall 0.952, F1 0.737, ROC-AUC 0.803, PR-AUC 0.794, Cost 895                  | Very close to HELOC winner, but still not a universal winner.                                                |
| Optuna / CV status                       | Taiwan 100/100 trials; HELOC 50/50 trials; both use 5x3 repeated stratified CV.                            | Cross-validation was used for hyperparameter optimization, not as a claim that every final table is CV-only. |
| Reproducibility                          | pytest: 39 passed.                                                                                         | Current artifact set is internally testable.                                                                 |

**Important transparency note:** These results do not prove that SCRE-Credit dominates all individual models. The strongest safe claim is that SCRE-Credit is a competitive reliability-aware framework, while CatBoost wins Taiwan expected cost and Scorecard wins HELOC expected cost.

**You can access all my results and code via my GitHub link;**  
[godsoftware/Data Mining Project](https://github.com/godsoftware/Data-Mining-Project)

#### Additional Evidence: Optuna / Cross-Validation Status

| Dataset | Status    | Completed Trials | CV Policy | Model Search Space          | Best Model | Best CV PR-AUC | Validation PR-AUC |
|---------|-----------|------------------|-----------|-----------------------------|------------|----------------|-------------------|
| heloc   | completed | 50               | 5x3       | catboost, lightgbm, xgboost | xgboost    | 0.7982         | 0.7968            |

|        |           |     |     |                                   |          |        |        |
|--------|-----------|-----|-----|-----------------------------------|----------|--------|--------|
| taiwan | completed | 100 | 5x3 | catboost,<br>lightgbm,<br>xgboost | catboost | 0.5684 | 0.5624 |
|--------|-----------|-----|-----|-----------------------------------|----------|--------|--------|

### Additional Evidence: Final Test Leaderboard Summary

| Dataset | Rank | Model              | Threshold | Accuracy | Precision | Recall | F1    | ROC - AUC | PR-AUC | Cost |
|---------|------|--------------------|-----------|----------|-----------|--------|-------|-----------|--------|------|
| Taiwan  | 1    | CatBoost           | 0.15      | 0.627    | 0.351     | 0.809  | 0.490 | 0.785     | 0.564  | 3247 |
| Taiwan  | 2    | Monotonic XGBoost  | 0.15      | 0.628    | 0.351     | 0.803  | 0.488 | 0.781     | 0.557  | 3280 |
| Taiwan  | 3    | XGBoost            | 0.14      | 0.598    | 0.336     | 0.836  | 0.479 | 0.783     | 0.562  | 3283 |
| Taiwan  | 4    | Monotonic LightGBM | 0.18      | 0.698    | 0.399     | 0.719  | 0.513 | 0.780     | 0.555  | 3304 |
| Taiwan  | 5    | LightGBM           | 0.17      | 0.669    | 0.375     | 0.746  | 0.499 | 0.778     | 0.555  | 3332 |
| Taiwan  | 6    | Old SCRE-Credit    | 0.16      | 0.654    | 0.364     | 0.756  | 0.492 | 0.782     | 0.544  | 3370 |
| HELOC   | 1    | Scorecard / WOE    | 0.16      | 0.587    | 0.559     | 0.982  | 0.712 | 0.801     | 0.804  | 892  |
| HELOC   | 2    | Old SCRE-Credit    | 0.17      | 0.646    | 0.601     | 0.952  | 0.737 | 0.803     | 0.794  | 895  |
| HELOC   | 3    | CatBoost           | 0.14      | 0.586    | 0.558     | 0.981  | 0.711 | 0.804     | 0.812  | 898  |
| HELOC   | 4    | Monotonic XGBoost  | 0.14      | 0.573    | 0.550     | 0.985  | 0.706 | 0.797     | 0.801  | 903  |

|        |   |                    |      |       |       |       |       |       |       |     |
|--------|---|--------------------|------|-------|-------|-------|-------|-------|-------|-----|
| HELO C | 5 | XGBoost            | 0.19 | 0.619 | 0.581 | 0.963 | 0.725 | 0.797 | 0.797 | 904 |
| HELO C | 6 | Monotonic LightGBM | 0.18 | 0.567 | 0.547 | 0.987 | 0.704 | 0.792 | 0.795 | 907 |

## SECTION 2: Preliminary Results & Literature Comparison

Report your current numerical results and compare them against the papers from your proposal. Fill in the table below.

| Model / Paper                 | Method                                | Accuracy | F1-Score | AUC-ROC | Other Metric                         | Best?   | Notes                                                                                                        |
|-------------------------------|---------------------------------------|----------|----------|---------|--------------------------------------|---------|--------------------------------------------------------------------------------------------------------------|
| Alam et al. (2020)            | GBDT + K-means SMOTE on Taiwan        | ~0.887   | NR       | NR      | Reported high accuracy               | No      | Not directly comparable until split/CV protocol is matched.                                                  |
| K-means SMOTE + BP NN (2021)  | Class imbalance handling + neural net | NR       | NR       | 0.929   | AUC improved 0.765 to 0.929          | No      | Very strong AUC claim; must be checked for protocol differences before comparing.                            |
| Interpretable DL + XAI (2023) | Deep model with XAI                   | 0.835    | NR       | NR      | Sensitivity 0.882, specificity 0.988 | No      | Current work is not claiming to exceed it.                                                                   |
| DNN empirical study (2025)    | Deep neural network on Taiwan         | 0.818    | 0.482    | 0.770   | G-mean 0.603                         | Partial | Closest to current Taiwan metrics; my F1 is similar but accuracy is lower at cost-sensitive thresholds.      |
| Lin & Wang (2025)             | 100 XGBoost seeds + SHAP stability    | 0.795    | NR       | 0.777   | Kendall W around 0.98                | Context | Relevant to explanation-stability framing, but not direct evidence for CatBoost or SCORE ensemble stability. |

|                           |                                                  |       |       |       |                            |                    |                                                                               |
|---------------------------|--------------------------------------------------|-------|-------|-------|----------------------------|--------------------|-------------------------------------------------------------------------------|
| MY best operational model | CatBoost + isotonic calibration + cost threshold | 0.628 | 0.490 | 0.785 | PR-AUC 0.564;<br>Cost 3247 | Objective-specific | Best current Taiwan cost result, but precision=0.351 is a serious limitation. |
| MY proposed framework     | SCRE-Credit reliability-aware ensemble           | 0.654 | 0.492 | 0.782 | PR-AUC 0.544;<br>Cost 3370 | No                 | Competitive framework, but not the best Taiwan cost model.                    |

## SECTION 2b: Results Discussion

**Q1:** The current Taiwan results are realistic but not state-of-the-art. The best cost model has ROC-AUC 0.785 and PR-AUC 0.564, but its accuracy is only 0.628 because threshold 0.15 is very aggressive.

**Q2:** The low F1 is mainly caused by low precision. The model catches many defaults, but it also flags many non-default customers. Therefore, high recall is purchased with too many false positives.

**Q3:** The results are not suspiciously high. That is good for leakage control, but it also means I should not overstate the model. The current system is better framed as screening/manual-review support than as an automatic lending decision system.

**Q4:** The expected contribution is not raw accuracy. The defensible contribution is a decision-aware framework that evaluates discrimination, calibration, cost, explanation stability context, ablation, and external validation.

## SECTION 3: Challenges & Problems Encountered

Describe any problems you faced this week. Be specific about error messages, unexpected results, or conceptual difficulties.

| Problem                                  | What I Tried / Found                                                                                                                                                | Status (Solved / Ongoing)                                       |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Low F1 and low apparent accuracy         | The best Taiwan operational threshold is 0.15. This creates high recall but too many false positives, so precision is only 0.351 and accuracy drops to 0.628.       | Ongoing - next week will test precision-constrained thresholds. |
| SCRE-Credit does not dominate all models | The claim-control matrix shows CatBoost wins Taiwan expected cost and Scorecard wins HELOC expected cost.                                                           | Solved by no-overclaim wording.                                 |
| HELOC interpretation risk                | The same Taiwan-trained model is not directly applied to HELOC because feature schemas differ. The result supports pipeline portability, not direct model transfer. | Solved; wording corrected.                                      |

|                                  |                                                                                                                                |                                                                   |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Explanation-stability limitation | The 50-seed SHAP stability protocol is directly computed for LightGBM, not for CatBoost, Scorecard, or SCRE ensemble variants. | <b>Ongoing - direct final-model stability can be future work.</b> |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|

## SECTION 4: Plan for Next Week

List 3-5 specific, measurable tasks you will complete by next Tuesday. These will be checked against your next report.

| # | Planned Task (be specific)                                                | Expected Deliverable                                                                                                                                                         |
|---|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Precision-constrained threshold search.                                   | Constraint table for loose, balanced, strong, conservative, specificity-aware, FP-budget, and hybrid threshold policies.                                                     |
| 2 | Cost matrix sensitivity for FN/FP scenarios 5/1, 5/2, 3/2, 2/2, and 10/1. | Table showing whether CatBoost, SCRE-Credit, Scorecard, XGBoost, or LightGBM wins under each cost assumption.                                                                |
| 3 | Decision curve / net benefit analysis.                                    | Net-benefit curves comparing treat-all, treat-none, CatBoost, SCRE-Credit, Scorecard, and XGBoost over threshold ranges.                                                     |
| 4 | Training-level imbalance tuning.                                          | CatBoost auto_class_weights, LightGBM scale_pos_weight/is_unbalance/class_weight, and XGBoost scale_pos_weight/max_delta_step/sample_weight results followed by calibration. |
| 5 | Realistic manual-review band optimization.                                | Three-way decision table with manual_review_rate <= 25%, high_risk_precision >= 0.45, low_risk_default_rate <= 0.10, and captured_defaults_high_risk >= 0.60.                |

## SECTION 5: Self-Assessment

| Question                                                           | Your Answer                                                                                                                                              |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overall project completion (%)                                     | About 85% for the experimental pipeline. The remaining work is mainly decision-policy refinement, not adding more unrelated models.                      |
| Did you complete last week's planned tasks? (Yes / Partially / No) | Yes. HELOC Optuna, SHAP stability evidence, SCRE ablation, calibration/final leaderboards, and claim-control outputs are now present.                    |
| If not, why?                                                       | The main remaining problem is not missing computation; it is that the current cost-minimizing threshold produces low precision and many false positives. |

|                                                        |                                                                                                                                                                                          |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Are you on track to finish the project on time?        | Yes, if next week focuses on threshold constraints, cost sensitivity, decision curves, imbalance tuning, and realistic manual-review policies.                                           |
| Do you need help from the instructor? If yes, on what? | Feedback would be useful on whether the final report should frame the contribution as cost-sensitive screening, precision-constrained lending policy, or manual-review decision support. |

## SUGGESTED WEEKLY MILESTONES

Use this as a guide for pacing your project. We assume you have already done 1-5. Your weekly reports should roughly follow this timeline.

| Week | Focus Area                    | Expected Tasks                                                                                      | Expected Deliverables                                                     |
|------|-------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1    | Data Loading & EDA            | Load dataset, check shape/types/missing values, compute basic statistics, create distribution plots | EDA summary table, 2-3 distribution plots, dataset description            |
| 2    | Data Cleaning & Preprocessing | Handle missing values, remove duplicates, filter outliers, encode categoricals, apply scaling       | Clean dataset stats, before/after comparison, preprocessing pipeline code |
| 3    | Feature Engineering           | Create domain-specific features, feature selection, create final feature set                        | Feature importance ranking, selected feature list with justification      |
| 4    | Baseline Model                | Train 2-3 baseline models with default parameters, evaluate with proper CV                          | Baseline comparison table with 3+ metrics, confusion matrix               |
| 5    | Imbalance Handling & Ablation | Compare SMOTE/ADASYN/cost-sensitive, test feature set variants                                      | Comparison table: original vs. engineered features, sampling strategies   |
| 6    | Proposed Model Implementation | Implement proposed methodology                                                                      | Proposed model results vs. baseline vs. literature comparison table       |
| 7    | Hyperparameter Optimization   | Run Optuna/GridSearch/Bayesian optimization                                                         | Optimized hyperparameters, before/after tuning comparison                 |
| 8    | Explainability & Analysis     | Run SHAP/LIME, create explanation plots, analyze feature importance, stability testing              | SHAP summary plot, LIME examples, feature importance comparison           |



|    |                            |                                                                                                |                                               |
|----|----------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 9  | Final Evaluation & Writing | Final hold-out test evaluation, bootstrap CIs, compile all results, start writing final report | Complete results table, all figures finalized |
| 10 | Final Report Submission    | Write discussion, conclusion, limitations, future work. Format and submit.                     | Final PDF report                              |

## WEEKLY REPORT GRADING CRITERIA

Each weekly report will be assessed on the following criteria:

| Criterion             | Weight | Description                                                                                |
|-----------------------|--------|--------------------------------------------------------------------------------------------|
| Progress Evidence     | 30%    | Did the student make tangible progress? Is there concrete evidence?                        |
| Results & Comparison  | 30%    | Are numerical results reported? Are they compared against literature?                      |
| Critical Thinking     | 20%    | Does the student explain why results are better/worse? Are problems analyzed thoughtfully? |
| Planning & Commitment | 20%    | Is the next-week plan specific and measurable?                                             |

### IMPORTANT WARNINGS

- Reports with NO numerical results or evidence will receive 0 for 'Progress Evidence'.
- Submitting the same content two weeks in a row will receive 0 for 'Progress Evidence'.
- If results exceed the literature by a suspiciously large margin, you must explain why. Unexplained perfect scores will be flagged.
- AI-generated text is not a substitute for your own analysis. Show YOUR understanding of YOUR results.
- Late submissions: same policy as the project proposal (up to 1 hour = 25% penalty, after 1 hour = not accepted).